

TEMPORARY RIVERS: LINKING ECOHYDROLOGY, ECOLOGICAL QUALITY AND RECONCILIATION ECOLOGY

A. H. ARTHINGTON^a, J. M. BERNARDO^b AND M. ILHÉU^{b,c,*}^a Australian Rivers Institute, Griffith University, Brisbane, Queensland, Australia^b Departamento de Paisagem, Ambiente e Ordenamento, Escola de Ciências e Tecnologia, Universidade de Évora, Évora, Portugal^c ICAAM—Instituto de Ciências Agrárias e Ambientais Mediterrânicas, Universidade de Évora, Nossa Senhora da Tourega, Portugal

ABSTRACT

Temporary streams and rivers, also referred to as intermittent, are defined as waterways that cease to flow at some points in space and time along their course. They are shaped by alternating wet and dry periods over annual and inter-annual cycles, making them one of the most dynamic freshwater ecosystems. These distinctive systems represent a substantial proportion of the total number, length and discharge of the global river network and are expected to become more widespread and face increasing pressures in many regions as a result of human activities and climate change. This collection of papers arose from the Conference on 'Ecohydrology and Ecological Quality in Temporary Rivers' held at the University of Evora, Portugal, 12–14 September 2012. The primary objectives of the meeting were to bring together researchers and expertise from a range of temporary ecosystems, from small temporary Mediterranean and Californian streams to intermittent rivers from semi-arid and arid zones. A common thread through the discussions was the assessment of ecological conditions within these dynamic ecosystems. To undertake a 'condition assessment' is challenging because of the need to differentiate between biotic responses to anthropogenic pressures and natural variability associated with the hydro-climatology and disturbance regime typical of these systems, that is, drought, drying and flooding. The management of degraded intermittent streams and rivers presents many new challenges, not least how to reconcile ecological and societal goals for emerging 'novel' ecosystems. Reconciliation ecology presents a realistic way of managing novel ecosystems given their value and desirable ecological services to society, rather than trying to restore to an original ecological state that may not be sustainable or economically possible. Copyright © 2014 John Wiley & Sons, Ltd.

KEY WORDS: temporary streams; intermittent rivers; ecological assessment; reconciliation ecology; water management

Received 16 July 2014; Accepted 23 July 2014

INTRODUCTION

Temporary running waters are the dominant freshwater type in Mediterranean, semi-arid and arid climate regions and are also common in many other regions, representing a substantial proportion of the total number, length and discharge of the global river network (Tooth, 2000; Larned *et al.*, 2010; Datry *et al.*, 2014).

Temporary streams and rivers, also referred as intermittent, are defined as waterways that cease to flow at some points in space and time along their course (Steward *et al.*, 2012; Acuña *et al.*, 2014). They are shaped by alternating wet and dry periods over annual and inter-annual cycles, making them one of the most dynamic freshwater ecosystems.

Because of the general nature of the definition, temporary or intermittent rivers are usually taken as a broad type encompassing very different ecosystems. With the proliferation of terms, a certain confusion has arisen over the use

of the terms temporary, intermittent, seasonal, periodic, episodic, and ephemeral, which are sometimes used synonymously. There is a considerable array of different definitions for running waters where loss of flow is a characteristic feature (e.g. Comin and Williams, 1994; Uys and O'Keefe, 1997; Boulton *et al.*, 2000; Williams, 2006). In addition to the hydrological regime, quantified by the timing, duration and predictability of flow, the availability of water and habitat throughout the year as well as the loss of hydrological continuity over both space and time should also be taken into account in the classification of temporary running waters. In an attempt to address the perceived inconsistencies in the terminology applied to these lotic ecosystems, we distinguish (i) temporary or intermittent—those in which the seasonal loss of flow and hydrological continuity results in high variability in the availability of water and habitat throughout the year and may partially dry at some times of the year (usually from weeks to a few months) on a more or less predictable basis, from (ii) episodic or ephemeral—those that flow and maintain water on a largely unpredictable basis, depending on precipitation events, and with hydrological continuity for only a short period of time (usually days to weeks).

*Correspondence to: M. Ilhéu, Departamento de Paisagem, Ambiente e Ordenamento, Escola de Ciências e Tecnologia, Universidade de Évora, Rua Romão Ramalho, 59, 7000-671 Évora, Portugal.
E-mail: milheu@uevora.pt

Aquatic habitats that are naturally intermittent are characterized by unique and diverse biological communities, distinctive biogeochemical processes and by the essential ecosystem goods and services they provide to societies. They are expected to become more widespread and more vulnerable in many regions as a result of anthropogenic pressures and climate change (Palmer *et al.*, 2008; Larned *et al.*, 2010; Sabater and Tockner, 2010). Given that temporary streams primarily occur in regions where water for human use is in short supply and demand is growing, intermittent riverine ecosystems are increasingly being modified. In the last few decades, growing anthropogenic-induced disturbances combined with strong seasonal and inter-annual flow variability, aggravated by potential climate change, has led to a reduction in the ecological quality of many intermittent rivers (Palmer *et al.*, 2008; Brooks, 2009; Matono *et al.*, 2012; Rosado *et al.*, 2012). Degradation of water quality (mainly eutrophication) caused by organic and nutrient inputs and hydro-morphological disturbances caused by land use intensification, water abstraction, river impoundments and channel diversions represent some of the most serious pressures upon temporary freshwaters, resulting in biodiversity losses and reduction in biotic integrity (e.g. Batalla *et al.*, 2004; Dodds *et al.*, 2004; Murchie *et al.*, 2008; Benejam *et al.*, 2009; Matono *et al.*, 2012). In regions subjected to high water resource and land use demands, such as the fertile valleys of Catalonia (Spain) and California (USA), many small lowland temporary streams have disappeared from the landscape as a result of over-exploitation, earthmoving, erosion and sedimentation; while others have become perennial because of flow regulation to supply water for irrigation (Batalla *et al.*, 2004; Kondolf and Batalla, 2005).

The intermittent character of stream flows and the annual variation of temperature strongly shape the environmental template to which the biota of intermittent waterways is subjected. Environmental conditions are particularly harsh for aquatic organisms during the dry period, when surface flow ceases and rivers form long dry reaches between shrinking isolated pools. During dry periods, the natural disturbance associated with the switch from lotic to lentic conditions, often combined with elevated temperatures, causes a sharp decline in environmental quality, with major effects on biotic assemblage structure and dynamics (Ilhéu, 2004; Magalhães *et al.*, 2007; Arthington *et al.*, 2010; Lake, 2011). As the biota has evolved in this dynamic and intermittently extreme environment, species exhibit resistance and resilience traits and frequently display high levels of endemism, making these communities biodiversity hot spots (Williams, 1998; Gasith and Resh, 1999; Dieterich and Anderson, 2000; Meyer *et al.*, 2007; McDonough *et al.*, 2011).

The dynamic characteristics of intermittent rivers present significant challenges for the assessment of ecological

conditions and potentially affect the accuracy of 'river health' monitoring results. The ebb and flow of water in channels and floodplains creates a heterogeneous matrix of aquatic, semi-aquatic and terrestrial habitat patches along intermittent streams and rivers. The availability of habitats and their suitability for completion of different species' life history stages may fluctuate dramatically. As a result, species assemblages also tend to display high spatial and temporal variability, both intra-annually and inter-annually (Boulton and Lake, 1992; Bernardo *et al.*, 2003; Stromberg *et al.*, 2005; Marshall *et al.*, 2006; Magalhães *et al.*, 2007). Condition assessments are therefore challenged to differentiate between biotic responses to imposed anthropogenic pressures and variations associated with the significant natural hydrological variability typical of these systems, that is, drought, drying and flooding (Sheldon, 2005; Matono *et al.*, 2012). Furthermore, the seasonal flow reduction during annual dry periods, and in dry years, typically magnifies the effect of anthropogenic pressures. This may lead to further reductions in water quality, habitat availability and diversity, and disrupt natural fluvial habitat connectivity. Following years with low precipitation and/or exceptionally hot, dry and long summers, lengthy reaches of dry riverbed may dramatically separate the remaining pool refuges (Bernardo and Alves, 1999). Under these conditions, aquatic biota may experience high mortality rates despite being resistant to some level of environmental stress, and the subsequent recolonization of the flowing river network may take longer or be less extensive than during wetter years, especially for fish (Kerezy *et al.*, 2013).

The volume and temporal variability of flow, water quality, habitat availability, quality, connectivity and persistence, food resources and biotic interactions during harsh dry periods are all critical factors influencing the biodiversity of intermittent riverine ecosystems and resistance and resilience to human pressures. The monitoring, management and conservation of freshwater biota in intermittent rivers therefore require a comprehensive approach involving recognition of their natural ecohydrology as well as responses to imposed human pressures. Understanding the relative importance and interactions of natural and human-induced pressures on intermittent freshwater systems is critical and will require the development and refinement of biological indicators, specific tools and experiments to assess their ecological condition and to monitor their long-term trends (Sheldon, 2005; Townsend *et al.*, 2008). Research efforts in these areas have commenced in some countries (e.g. Australia and parts of Europe), but the imperative remains to explore new methods and tools and urgently promote their application as intermittent streams and rivers are subject to increasing pressure from human activities and environmental change.

This Special Issue of River Research and Applications draws together papers from the Conference on 'Ecohydrology and Ecological Quality in Temporary Rivers' held at the University of Evora, Portugal, 12–14 September 2012. The subject matter of the contributions spans patterns of biodiversity and ecohydrology, impacts of impoundment and flow regulation, assemblage composition and biotic interactions involving alien species, ecological condition ('health') assessment, and the concept of reconciliation ecology as a strategy to actively guide the management of degraded rivers towards novel ecosystems with desirable features, including supporting native species. Intermittent rivers present a significant 'challenge for freshwater ecology' (Datry *et al.*, 2014). We hope that the papers in this Special Issue and the following synthesis will foster greater interest on this distinctive type of river system and help resolve some the challenges they present for scientists, water resource and land use managers, and wider society in general.

SYNTHESIS OF PAPERS

A good place to start this synthesis is to ask how well we understand patterns of biodiversity and autecology in naturally intermittent streams and rivers and to identify key knowledge gaps that constrain the identification of sensitive biological indicators. One of the lesser known aquatic groups in lotic ecosystems is benthic diatoms. Their use as indicators is currently limited when assessing water quality as required under the European Water Framework Directive (WFD, Directive 2000/60/EC, 2000). Diatom taxon richness is higher in temporary Mediterranean watercourses than in permanently flowing systems, providing a strong rationale for further research to assess their utility as biological indicators and support the conservation of biodiversity in headwater streams in particular (Novais *et al.*, in press).

The WFD requires standardised sampling of taxonomic groups such as benthic macroinvertebrates to support stream condition assessments. Most WFD macroinvertebrate protocols are based on information taken from a composite sample drawn from major habitat types in proportion to their areal representation in a study site. Leitão *et al.* (2014) demonstrates that this composite sampling method probably omits taxa occupying microhabitats occurring in very low proportions in the system overall. They recommend a more refined 'habitat sampling protocol' to achieve better estimates of macroinvertebrate community composition and more realistic river condition 'metrics' in temporary streams.

The WFD also requires the use of fish to assess river ecosystem condition. However, intermittent Mediterranean streams have low local fish species richness (but with a high

proportion of endemic taxa), and assemblages are dominated by generalist taxa. In addition, multiple natural and anthropogenic stressors complicate the identification of fish community metrics responsive to specific human pressures (Hermoso *et al.*, 2010). Nevertheless, Godinho *et al.* (2014) identified a series of fish community metrics clearly indicating changes associated with impoundment and irrigation intensification within intermittent streams of south-western Iberia. The proportion of less tolerant species and native invertivores was found to decrease with increasing levels of irrigation, while most introduced species increased in their relative abundance with an increasing area of irrigated land in the catchment. The Alqueva reservoir allowed non-native species to thrive and probably facilitated their expansion into nearby intermittent habitats where they impacted native fish assemblages through predation and direct competition (Clavero and Hermoso, 2011).

Ideal indicators for assessing river condition are those that are easy to measure, relevant to river health and can deliver early warning signals of pressure or diagnose causes of poor ecological conditions (Boulton, 1999). In the intermittent floodplain rivers of semi-arid Australia, Balcombe *et al.* (2014) identified simple, non-destructive metrics based on fish weight and length (reflecting body condition and juvenile recruitment) related to natural antecedent flow conditions in rivers with contrasting flow variability and predictability. They suggest that it may be possible to identify suites of ecohydrological relationships for intermittent rivers with contrasting hydrological regimes, and from such models predict how the biota of each river flow type will respond to flow regime modifications or managed environmental flows. However, to be more widely useful, such metrics and relationships will require calibration for species with different trait characteristics (e.g. reproductive tactics) in rivers of contrasting hydrological regime (refer to Poff *et al.*, 2010).

The persistence of fish populations in Mediterranean streams and arid-zone rivers is dependent upon residual dry season pools that act as ecological refugia for obligate aquatic species and provide a source of colonists into dewatered areas when flow events reconnect isolated habitats (Pires *et al.*, 2010; Sheldon *et al.*, 2010). The diverse movement behaviours of fish assemblages appear to be a critical factor in the recovery of intermittent river ecosystems following dry periods (e.g. Hodges and Magoulick, 2011; Kerezy *et al.*, 2013). Using mark-recapture techniques, Pires *et al.* (2014) provide novel insights into the movement dynamics (e.g. timing of movements and distance covered) of the endangered chub *Squalius torgalensis*, a cyprinid native to seasonally drying Mediterranean streams in south-west Portugal. Understanding movement behaviour is critical for predicting how fish will adapt in rivers where human interventions (e.g. barriers

and river flow regulation) or more extreme climatic events, such as prolonged drought, disrupt connectivity pathways and exacerbate the effect of habitat fragmentation.

Dams, irrigation schemes and agricultural intensification typically impose a suite of stresses on riverine ecosystems that may severely impact biotic assemblages (Ormerod *et al.*, 2010). Multiple interacting natural and human-induced disturbances may compromise biotic assemblage stability (i.e. the relative constancy of species abundance over time) and persistence. However, predicting which assemblages are most vulnerable is challenging. Matono *et al.* (2014) successfully distinguished fish assemblage patterns associated with land use intensification from those related to large-scale and regional natural environmental drivers in Portuguese temporary streams. Their study highlights the vulnerability of fish assemblages in upstream sectors of lowland streams to anthropogenic disturbances, particularly land use intensification and riparian habitat degradation. Mitigation strategies to strengthen resilience and reduce the impacts of agricultural intensification are highlighted.

Interactions of stressors associated with river impoundment and flow regulation typically alter native riverine fish assemblages and simultaneously promote the establishment and spread of non-native fishes, invertebrates and plants (Bunn and Arthington, 2002; Ribeiro and Leunda, 2012). Studies to test the interaction of these two disruptive factors (flow regulation and non-native species) and their synergistic effects are rare, especially their effects on higher trophic levels of stream food webs and trophic ecology more widely. Flow regulation and non-native species have increased prey availability for the Eurasian otter (*Lutra lutra*) in regulated Mediterranean watercourses. Bueno-Enciso *et al.* (2014) demonstrate a shift in the natural diet of Eurasian otter from a dominance of native prey to introduced species, particularly red-swamp crayfish (*Procambarus clarkii*) and pumpkinseed sunfish (*Lepomis gibbosus*). The paper proposes management options to protect the high native biodiversity of temporary Mediterranean streams and achieve overall conservation benefits, including minimising impacts on food webs.

Non-native crayfish have resulted in the displacement of numerous native crayfish populations throughout North America and Europe; yet, the consequences of species displacements upon community structure and ecosystem processes are poorly understood (Lodge *et al.*, 2012). In the next contribution, stream mesocosm experiments illustrate that native and invasive crayfish species may be largely ecologically redundant in their effects on ecosystem processes (leaf breakdown, community metabolism, periphyton, sediment) and chironomid densities (Magoulick, 2014). Factorial experiments also measured the effects of simulated drought on these processes; however, there was no

interaction between drought and crayfish presence on ecosystem variables. Magoulick (2014) suggests that there is a need for more complex mesocosm experiments encompassing a wider range of environmental conditions and food items to examine the effects of drought, invasive species and ecological redundancy.

The European WFD (Directive 2000/60/EC, 2000) requires an assessment of the ecological status of water bodies and achievement of 'good ecological status' for all surface water bodies by 2015. This ambitious policy has stimulated numerous investigations to test the relevance of methods developed for perennial streams, recognising their limitations for assessment of intermittent (temporary) streams. The Mediterranean Intermittent River ManAgement (MIRAGE) project has developed a variety of tools to assess the hydrological regime type and status, water chemistry and the ecological condition of intermittent streams (Prat *et al.*, 2014 and references therein). For example, the MIRAGE project is investigating terrestrial invertebrates as a means to assess ecological status when the stream channel has no surface water. This approach anticipates that a distinctive community of terrestrial invertebrates inhabits the riverbed during dry phases (Steward *et al.*, 2011). Measurements of both structural (e.g. taxonomic diversity, community composition of diatoms, macroinvertebrates and fish) and functional aspects of ecosystem processes (e.g. litter breakdown) are included in the MIRAGE toolbox, as together they can provide a more comprehensive assessment of stream ecological status than structural measures alone (Young *et al.*, 2008; Bunn *et al.*, 2010).

Temporary streams in California, the Iberian Peninsula and other Mediterranean climate regions are so modified by human actions and invasive species that they now support quite different or 'novel' ecosystems (Hobbs *et al.*, 2009) dominated by new combinations of species in highly altered habitats (Hermoso *et al.*, 2011). The emergence of novel ecosystems is expected to increase in many regions as a result of multiple human pressures, further species invasions and the likely effects of climate change (Palmer *et al.*, 2008; Sabater and Tockner, 2010). Restoration of such rivers to a previous state is idealistic and usually unachievable given the magnitude of change to hydrology, geomorphology, biotic composition and ecosystem processes (Seastedt *et al.*, 2008; Dufour and Piégay, 2009; Acreman *et al.*, in press). As Moyle (2014) eloquently explains, reconciliation ecology offers a realistic way of managing novel ecosystems for their values and desirable services to society, which may include physical characteristics, ecological processes and native species. Case studies from California's temporary streams illustrate the scientific and social principles for reconciling novel ecosystems and practices that have succeeded in conserving some native

species even though non-native invaders still persist in highly regulated and modified river reaches. Reconciling particular novel ecosystems may be costly, and the risks substantial, as management actions may result in unexpected outcomes, ecological surprises and more novelty.

CONCLUSIONS

Temporary streams and rivers may become one of the most widespread and distinctive ecosystem types in the global river network. These dynamic systems present numerous challenges for scientific research, monitoring, management, societal valuation and setting of policy goals. Papers in this Special Issue embrace these challenges with vigour and innovation through consideration of patterns of biodiversity and ecohydrology (Balcombe *et al.*, 2014; Leitão *et al.*, 2014; Novais *et al.*, in press; Pires *et al.*, 2014), impacts of impoundment, flow regulation and agricultural intensification (Godinho *et al.*, 2014; Matono *et al.*, 2014), community composition and biotic interactions involving alien species (Bueno-Enciso *et al.*, 2014; Magoulick, 2014), ecological condition (health) assessment (Prat *et al.*, 2014) and the concept and practice of reconciliation ecology (Moyle, 2014) as a way of managing novel ecosystems for their values and desirable ecological services to society, rather than trying to restore an unachievable original ecological state.

A new science of temporary (intermittent) streams and river ecology and management is emerging from studies in Mediterranean and other water-stressed areas of the world. The way forward for temporary river science is challenging, as temporary lotic systems are expected to become more widespread and face increasing pressures in many regions as a result of human activities and climate change. Greater investment in temporary river science is timely, essential and certain to be richly rewarding.

ACKNOWLEDGEMENTS

The Guest Editors wish to thank the authors for their contributions to this special issue. We would like to thank the numerous manuscript reviewers for their hard work in enhancing the quality and clarity of the papers included. We are very grateful to Paul Wood for his work involved in the preparation of this special issue of 'River Research and Applications' and Geoff Petts for allowing us to prepare it for the journal. This Special Issue includes papers presented at the TEMPRIV Conference held at the University of Evora, Portugal, 12–14 September 2012 and organized within the research project 'Climate change and fish communities of Mediterranean-type streams. Potential impact on the bio-integrity and implications on the ecological status assessment' funded by the Portuguese Agency for

Science and Technology, Fundação para a Ciência e Tecnologia, (Project PTDC/AAC-AMB/102541/2008).

REFERENCES

- Acreman MC, Arthington AH, Colloff MC, Couch C, Crossman N, Dyer F, Overton I, Pollino CA, Stewardson MJ, Young W. in press. Environmental flows for natural, hybrid and novel riverine ecosystems in a changing world. *Frontiers in Ecology and the Environment*.
- Acuña V, Datry T, Marshall J, Barceló D, Dahm CN, Ginebreda A, McGregor G, Sabater S, Tockner K, Palmer MA. 2014. Why should we care about temporary waterways? *Science* **343**: 1080–1081. DOI: 10.1126/science.1246666
- Arthington AH, Olden JD, Balcombe SR, Thoms MC. 2010. Multi-scale environmental factors explain fish losses and refuge quality in drying waterholes of Cooper Creek, an Australian arid-zone river. *Marine and Freshwater Research* **61**: 842–856. DOI: 10.1071/MF09096
- Balcombe SR, Arthington AH, Sternberg D. 2014. Metrics of fish body condition and recruitment success reflect flow magnitude and timing in two dryland rivers. *River Research and Applications* 7 August 2014. DOI: 10.1002/rra.2797
- Batalla RJ, Gómez CM, Kondolf GM. 2004. Reservoir-induced hydrological changes in the Ebro River basin (NE Spain). *Journal of Hydrology* **290**: 117–136. DOI: 10.1016/j.jhydrol.2003.12.002
- Benejam L, Angermeier P, Munne A, Garcia-BE. 2009. Assessing effects of water abstraction on fish assemblages in Mediterranean streams. *Freshwater Biology* **55**: 628–642. DOI: 10.1111/j.1365-2427.2009.02299.x
- Bernardo JM, Alves MH. 1999. New perspectives for ecological flow determination in semi-arid regions: a preliminary approach. *Regulated Rivers: Research and Management* **15**: 221–229. DOI: 10.1002/(SICI)1099-1646(199901/06)15:1/3<221::AID-RRR537>3.0.CO;2-A
- Bernardo JM, Ilhéu M, Matono P, Costa AM. 2003. Interannual variation of fish assemblage structure in a Mediterranean river: implications of streamflow on the dominance of native or exotic species. *River Research and Applications* **19**: 1–12. DOI: 10.1002/rra.726
- Boulton AJ. 1999. An overview of river health assessment: philosophies, practice, problems and prognosis. *Freshwater Biology* **41**: 469–480.
- Boulton AJ, Lake PS. 1992. The ecology of two intermittent streams in Victoria, Australia. III. Temporal changes in faunal composition. *Freshwater Biology* **27**: 123–138. DOI: 10.1111/j.1365-2427.1992.tb00528.x
- Boulton AJ, Sheldon F, Thoms MC, Stanley EH. 2000. Problems and constraints in managing rivers with variable flow regimes. In *Global Perspectives on River Conservation: Science, Policy and Practice*, Boon PJ, Davies BR, Petts GE (eds). John Wiley and Sons: London, 414–426.
- Brooks TT. 2009. Potential impacts of global climate change on the hydrology and ecology of ephemeral freshwater systems of the forest of the northeastern United States. *Climate Change* **95**: 469–483. DOI: 10.1007/s10584-008-9531-9
- Bueno-Enciso J, Diaz-Ruiz F, Almeida D, Ferreras P. 2014. Effects of flow regulation and non-native species on feeding habits of eurasian otter *Lutra lutra* in Mediterranean temporary rivers. *River Research and Applications*. DOI: 10.1002/rra.2736
- Bunn SE, Abal EG, Smith MJ, Choy SC, Fellows CS, Harch BD, Kennard MJ, Sheldon F. 2010. Integration of science and monitoring of river ecosystem health to guide investments in catchment protection and rehabilitation. *Freshwater Biology* **25**: 223–240. DOI: 10.1111/j.1365-2427.2009.02375.x
- Bunn SE, Arthington AH. 2002. Basic principles and ecological consequences of altered flow regimes for aquatic biodiversity. *Environmental Management* **30**: 492–507. DOI: 10.1007/s00267-002-2737-0

- Clavero M, Hermoso V. 2011. Reservoirs promote the taxonomic homogenization of fish communities within river basins. *Biodiversity and Conservation* **20**: 41–57. DOI: 10.1007/s10531-010-9945-3
- Comin FA, Williams WD. 1994. Parched continents: our common future? In *Limnology now: a paradigm of planetary problems*, Margalef R (ed). Elsevier: Amsterdam; 473–527.
- Datry T, Larned ST, Tockner K. 2014. Intermittent Rivers: A Challenge for Freshwater Ecology. *Bioscience Advance Access* January 2014. DOI: 10.1093/biosci/bit027
- Dieterich M, Anderson NH. 2000. The invertebrate fauna of summer-dry streams in western Oregon. *Archiv für Hydrobiologie* **147**: 273–295.
- Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000. Establishing a framework for community action in the field of water policy. *Official Journal of the European Communities* **L327**: 1–72.
- Dodds WK, Gido K, Whiles MR, Fritz KM, Matthews WJ. 2004. Life on the edge: the ecology of Great Plains prairie streams. *BioScience* **54**: 205–216.
- Dufour SH, Piégay H. 2009. From the myth of a lost paradise to targeted river restoration: forget natural references and focus on human benefits. *River Research and Applications* **25**: 568–581. DOI: 10.1002/rra.1239
- Gasith A, Resh VH. 1999. Streams in Mediterranean climate regions: abiotic influences and biotic responses to predictable seasonal events. *Annual Review of Ecology and Systematics* **30**: 51–81. DOI: 10.1146/annurev.ecolsys.30.1.51
- Godinho FN, Pinheiro PJ, Oliveira JM, Azedo R. 2014. Responses of intermittent stream fish assemblages to irrigation development. *River Research and Applications*, 17 March 2014. DOI: 10.1002/rra.2248
- Hermoso V, Clavero M, Blanco-Garrido F, Prenda J. 2010. Assessing the ecological status in species-poor systems: a fish-based index for Mediterranean rivers. *Ecological Indicators* **10**: 1152–1161. DOI: 10.1016/j.ecolind.2010.03.018
- Hermoso V, Clavero M, Blanco-Garrido F, Prenda J. 2011. Invasive species and habitat degradation in Iberian streams: an analysis of their role in freshwater fish diversity loss. *Ecological Applications* **21**: 175–188. DOI: 10.1890/09-2011.1
- Hobbs RJ, Higgs E, Harris JA. 2009. Novel ecosystems: implications for conservation and restoration. *Trends in Ecology and Evolution* **24**: 599–605. DOI: 10.1016/j.tree.2009.05.012
- Hodges SW, Magoulick DD. 2011. Refuge habitats for fishes during seasonal drying in an intermittent stream: Movement, survival and abundance of three minnow species. *Aquatic Sciences* **73**: 513–522.
- Ilhéu M. 2004. Patterns of habitat use by freshwater fishes in Mediterranean rivers. PhD Thesis, University of Evora.
- Kerezszy A, Balcombe SJ, Tischler M, Arthington AH. 2013. Fish movement strategies in an ephemeral river in the Simpson Desert, Australia. *Austral Ecology* **38**: 798–808. DOI: 10.1111/aec.12075
- Kondolf GM, Batalla RJ. 2005. Hydrologic effects of dams and water diversions on rivers in Mediterranean-climate regions: examples from California. In *Catchment Dynamics and River Processes: Mediterranean and Other Climate Regions*. Garcia C, Batalla RJ (eds). Elsevier B.V.: San Diego; 197–211.
- Lake PS. 2011. *Drought and aquatic ecosystems: Effects and responses*. Wiley-Blackwell: Oxford, England.
- Larned ST, Datry T, Arscott DB, Tockner K. 2010. Emerging concepts in temporary-river ecology. *Freshwater Biology* **55**: 717–738. DOI: 10.1111/j.1365-2427.2009.02322.x
- Leitão F, Hughes SJ, Maximo I, Atanasova N, Furtado A, Chicharo L. 2014. Habitat oriented sampling of macroinvertebrates affects the determination of ecological status in temporary Mediterranean river systems. *River Research and Applications*, 1 August 2014. DOI: 10.1002/rra.2800
- Lodge DM, Deines A, Gherardi F, Yeo DCJ, Arcella T, Baldrige AK, Barnes MA, Chadderton WL, Feder JL, Gantz CA, Howard GW, Jerde CL, Peters BW, Peters JA, Sargent LW, Turner CR, Wittmann ME, Zeng YW. 2012. Global introductions of crayfishes: Evaluating the impact of species invasions on ecosystem services. *Annual Review of Ecology, Evolution, and Systematics* **43**: 449–472. DOI: 10.1146/annurev-ecolsys-111511-103919
- Magalhães MF, Beja P, Schlosser IJ, Collares-Pereira MJ. 2007. Effects of multi-year droughts on fish assemblages of seasonally drying Mediterranean streams. *Freshwater Biology* **52**: 1494–1510. DOI: 10.1111/j.1365-2427.2007.01781.x
- Magoulick DD. 2014. Impacts of drought and crayfish invasion on stream ecosystem structure and function. *River Research and Applications*. DOI: 10.1002/rra.2747
- Marshall JC, Sheldon F, Thoms M, Choy S. 2006. The macroinvertebrate fauna of an Australian dryland river: spatial and temporal patterns and environmental relationships. *Marine and Freshwater Research* **57**: 61–74. DOI: 10.1071/MF05021
- Matono P, Bernardo JM, Costa AM, Ilhéu M. 2014. Fish response to human-induced pressures in temporary streams: the importance of environmental drivers. *River Research and Applications* DOI: 10.1002/rra.2780
- Matono P, Bernardo JM, Oberdorff T, Ilhéu M. 2012. Effects of hydrological variability on fish assemblages in small Mediterranean streams: implications for ecological assessment. *Ecological Indicators* **23**: 467–481. DOI: 10.1016/j.ecolind.2010.04.024
- McDonough OT, Hosen JD, Palmer MA. 2011. Temporary streams: the hydrology, geography, and ecology of non-perennially flowing waters. In *River Ecosystems: Dynamics, Management and Conservation*. Elliot HS, Martin LE (eds). Nova Science Publishers: Hauppauge, NY; 258–289.
- Meyer JL, Strayer DL, Wallace JB, Eggert SL, Helfman GS, Leonard NE. 2007. The contribution of headwater streams to biodiversity in river networks. *Journal of the American Water Resources Association* **43**: 86–103. DOI: 10.1111/j.1752-1688.2007.00008.x
- Moyle PB. 2014. Novel aquatic ecosystems: the new reality for streams in California and other Mediterranean climate regions. *River Research and Applications*. DOI: 10.1002/rra.2709
- Murchie KJ, Hair KPE, Pullen CE, Redpath TD, Stephens HR, Cooke SJ. 2008. Fish response to modified flow regimes in regulated rivers: Research methods, effects and opportunities. *River Research and Applications* **24**: 197–217. DOI: 10.1002/rra.1058
- Novais MH, Morais MM, Rosado J, Dias LS, Hoffmann L, Ector L. in press. Diatoms of temporary and permanent watercourses in Southern Europe (Portugal). *River Research and Applications*.
- Ormerod SJ, Dobson M, Hildrew AG, Townsend CR. 2010. Multiple stressors in freshwater ecosystems. *Freshwater Biology* **55**(Suppl 1): 1–4. DOI: 10.1111/j.1365-2427.2009.02395.x
- Palmer MA, Reidy Liermann CA, Nilsson C, Flörke M, Alcamo J, Lake PS, Bond N. 2008. Climate change and the world's river basins: anticipating management options. *Frontiers in Ecology and the Environment* **6**: 81–89. DOI: 10.1890/060148
- Pires DF, Beja P, Magalhães MF. 2014. Out of pools: movement patterns of Mediterranean stream fish in relation to dry season refugia. *River Research and Applications*. DOI: 10.1002/rra.2776
- Pires DF, Pires AM, Collares-Pereira MJ, Magalhães MF. 2010. Variation in fish assemblages across dry-season pools in a Mediterranean stream: effects of pool morphology, physicochemical factors and spatial context. *Ecology of Freshwater Fish* **19**: 74–86. DOI: 10.1111/j.1600-0633.2009.00391.x
- Poff NL, Richter BD, Arthington AH, Bunn SE, Naiman RJ, Kendy E, Acreman M, Apse C, Bledsoe BP, Freeman MC, Henriksen J, Jacobson RB, Kennen JG, Merritt DM, O'Keeffe JH, Olden JD, Rogers K, Tharme RE, Warne A. 2010. The ecological limits of hydrologic alteration

- (ELOHA): a new framework for developing regional environmental flow standards. *Freshwater Biology* **55**: 147–170. DOI: 10.1111/j.1365-2427.2009.02204.x
- Prat N, Gallart F, von Schiller D, Polesello S, García-Roger EM, Latron J, Rieradevall M, Llorens P, Barberá GG, Brito D, de Girolamo AM, Dieter D, Lo Porto A, Buffagni A, Erba S, Nikolaidis NP, Querner EP, Tournoud MG, Tzoraki O, Skoulikidis N, Gómez R, Sánchez-Montoya MM, Tockner K, Froebrich J. 2014. The MIRAGE toolbox: an integrated assessment tool for temporary streams. *River Research and Applications*. DOI: 10.1002/rra.2757
- Ribeiro F, Leunda PM. 2012. Non-native fish impacts on Mediterranean freshwater ecosystems: current knowledge and research needs. *Fisheries Management and Ecology*. DOI: 10.1111/j.1365-2400.2011.00842.x
- Rosado J, Morais M, Serafim A, Pedro A, Silva H, Potes M, Brito D, Salgado R, Neves R, Lillebø AI, Chambel A, Pires V, Pinto Gomes C, Pinto P. 2012. Key factors in the management and conservation of temporary mediterranean streams: A case study of the Pardiela River, Southern Portugal. In *River Conservation and Management*, Boon PJ, Raven PJ (eds). John Wiley & Sons, Ltd: Chichester, UK. DOI: 10.1002/9781119961819.ch22
- Sabater S, Tockner K. 2010. Effects of hydrologic alterations on the ecological quality of river ecosystems. In *Water Scarcity in the Mediterranean: Perspectives Under Global Change*, The Handbook of Environmental Chemistry 8, Sabater S, Barcelo D (eds). Springer Verlag: Berlin, Heidelberg. DOI: 10.1007/978-3-642-009_24
- Seastedt TR, Hobbs RJ, Suding KN. 2008. Management of novel ecosystems: are novel approaches required? *Frontiers in Ecology and Environment* **6**: 547–553. DOI: 10.1890/070046
- Sheldon F. 2005. Incorporating natural variability into the assessment of ecological health in Australian dryland rivers. *Hydrobiologia* **552**: 45–56. DOI: 10.1007/s10750-005-1504-7
- Sheldon F, Bunn SE, Hughes JM, Arthington AH, Balcombe SR, Fellows CS. 2010. Dryland river waterholes: A review of the ecological roles and threats to aquatic refugia in arid landscapes. *Marine and Freshwater Research* **61**: 885–895. DOI: 10.1071/MF09239
- Steward AL, Marshall JC, Sheldon F, Harch B, Choy S, Bunn SE, Tockner K. 2011. Terrestrial invertebrates of dry river beds are not simply subsets of riparian assemblages. *Aquatic Sciences* **73**: 551–566. DOI: 10.1007/s00027-011-0217-4
- Steward AL, von Schiller D, Tockner K, Marshall JC, Bunn SE. 2012. When the river runs dry: human and ecological values of dry riverbeds. *Frontiers in Ecology and the Environment* **10**: 202–209. DOI: 10.1890/110136
- Stromberg JC, Bagstad KJ, Leenhouts JM, Sharon JL, Makings E. 2005. Effects of stream flow intermittency on riparian vegetation of semiarid region river (San Pedro River, Arizona). *River Research and Applications* **21**: 925–938. DOI: 10.1002/rra.858
- Tooth S. 2000. Process, form and change in dryland rivers: a review of recent research. *Earth-Science Reviews* **51**: 67–107. DOI: 10.1016/S0012-8252(00)00014-3
- Townsend CR, Uhlmann SS, Matthaei CD. 2008. Individual and combined responses of stream ecosystems to multiple stressors. *Journal of Applied Ecology* **45**: 1810–1819. DOI: 10.1111/j.1365-2664.2008.01548.x
- Uys MC, O'Keeffe JH. 1997. Simple Words and Fuzzy Zones: Early Directions for Temporary River Research in South Africa. *Environmental Management* **21**: 517–531. DOI: 10.1007/s002679900047
- Williams DD. 1998. The role of dormancy in the evolution and structure of temporary water invertebrate communities. *Archiv für Hydrobiologie Special Issues Advances in Limnology* **52**: 109–124.
- Williams DD. 2006. *The Biology of Temporary Waters*. Oxford University Press: Oxford.
- Young RG, Matthaei CD, Townsend CR. 2008. Organic matter breakdown and ecosystem metabolism: functional indicators for assessing river ecosystem health. *Journal of the North American Benthological Society* **27**: 605–625. DOI: 10.1899/07-121.1